

Investigation into the prevalent CRISPR–Cas systems among the *Aeromonas* genus

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Abstract

Clustered regularly interspaced short palindromic repeats (CRISPR) is a prokaryotic adaptive immune system that checks invasion by mobile genetic elements through nuclease targeting. In this study, we investigated the occurrence, diversity, and features of the CRISPR system in the genus *Aeromonas* using bioinformatics tools. Only 13 out of 122 complete genomes (10.66%) of the genus *Aeromonas* from the NCBI GenBank database harbored the CRISPR system. The Type I–F system was the most prevalent CRISPR system among the Aeromonads, followed by the Type I–E system. Only one strain harbored a Type I–C CRISPR system. Among the Aeromonads, *Aeromonas caviae* (22.7%) and *Aeromonas veronii* (20%) had a higher prevalence rate of the complete CRISPR system. The analysis of direct repeat (DR) sequences showed that all could form stable RNA secondary structures. A phylogenetic tree generated for the Cas1 protein classified CRISPR subtypes into three distinct clusters. Among the 748 spacers investigated, 41.98% and 17.25% showed perfect homology to phage and plasmid sequences, respectively. Some arrays had duplicated spacers. The CRISPR loci are closely linked to antibiotic resistance genes in most strains. Collectively, our results would contribute to research on antibiotic resistance in the *Aeromonas* group, and provide new insights into the diversity and evolution of the CRISPR–Cas system.

KEYWORDS

Aeromonas, Cas proteins, CRISPR, direct repeat, spacer

1 | INTRODUCTION

The genus *Aeromonas* comprises Gram-negative bacteria that are ubiquitous in aquatic habitats and sediments [1]. Aeromonads are causative agents of diseases such as furunculosis and septicemia in fish [1]. In humans, *Aeromonas* species are associated with infections such as gastrointestinal infections, cellulitis, wound infections, septicemia, urinary tract infections, and hepatobiliary and ear infections [2]. Their ability to penetrate new niches and hosts and

establish themselves as pathogens comes from acquired horizontally transferred genes related to toxin production, biofilm formation, antibiotic resistance, and other virulence determinants [3]. The CRISPR (clustered, regularly interspaced, short palindromic repeats) system limits the horizontal transfer of genetic material. It acts as a prokaryotic immune system by limiting the establishment of plasmids and phage infections through the action of Cas (CRISPR-associated) proteins [4]. Typically, a CRISPR system consists of a CRISPR array, a series of *cas* genes, and a

Abbreviations: CRISPR, clustered regularly interspaced short palindromic repeats; NCBI, National Center for Biotechnology Information.

leader region [5]. The CRISPR array consists of identical repeats separated by unique spacers acquired from foreign genetic elements [6]. The AT-rich leader sequence serves as a promoter for the transcription of CRISPR genes [6]. Any invading foreign DNA that is complementary to the spacer is detected and degraded by Cas proteins [6]. In addition, it has also been associated with alternative roles such as regulation of virulence, genome evolution, and DNA repair [7]. To date, there are no comprehensive reports on the type and features of the complete CRISPR–Cas systems prevalent in the genus *Aeromonas*. Therefore, in this study, the publicly available complete genomes of the genus *Aeromonas* were screened for the presence of complete CRISPR loci. Furthermore, their features and potential functions were investigated using bioinformatics tools. Our results provide valuable insights into the existing diversity of the CRISPR–Cas immune system in the genus *Aeromonas*.

2 | MATERIALS AND METHODS

2.1 | Data sources

A data set of 122 complete genomes available for the genus *Aeromonas* as of February 25, 2021, was retrieved from the GenBank database (<https://www.ncbi.nlm.nih.gov/>).

The data set included complete genomes of *Aeromonas hydrophila* (33), *Aeromonas salmonicida* (16), *Aeromonas media* (10), *Aeromonas caviae* strains (23), *Aeromonas veronii* (21), *Aeromonas schubertii* (2), *Aeromonas jandaei* (4), *Aeromonas dhakensis* (1), *Aeromonas simiae* (1), *Aeromonas rivipollensis* (1), *Aeromonas encheilia* (1), *Aeromonas allosaccharophila* (1), and eight other *Aeromonas* spp. (1 each). The complete genomes were screened for the presence of complete CRISPR–Cas loci using the web tool CRISPRminer2 (<http://www.microbiome-bigdata.com/CRISPRminer2/index/predict>) (Figure 1). CRISPRminer2 is a comprehensive tool that uses a comparative genomics approach to identify and annotate CRISPR–Cas loci. This tool also helps with multiple detection options, including anti-CRISPR detection and annotation, self-targeting spacer search, repeat type identification, bacteria–phage interaction detection, and prophage detection [8]. Only “confirmed CRISPRs” identified by the CRISPRminer2 tool were selected for further analysis.

2.2 | Analysis of features of the complete CRISPR–Cas system

The PathoSystems Resource Integration Center (PATRIC) database provides genomic data and metadata for major bacterial lineages (<https://www.patricbrc.org>). The Compare Region Viewer tool of the PATRIC database

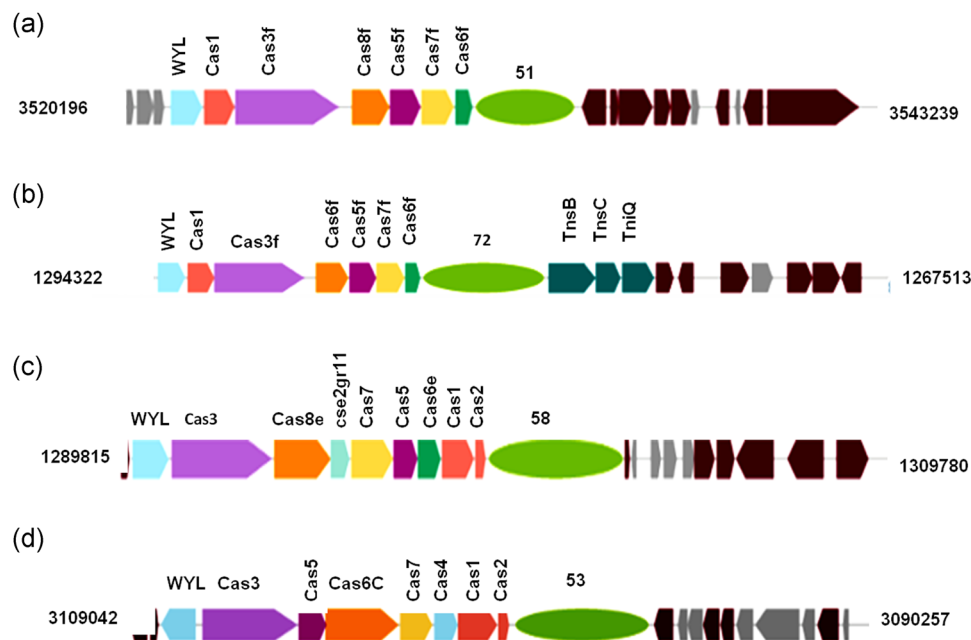


FIGURE 1 Architecture and organization of the CRISPR–Cas loci prevalent in the genus *Aeromonas* generated using the CRISPRminer2 tool. A representative of each type has been shown. (a) Type I–F CRISPR system; (b) Type I–F CRISPR associated with transposon genes; (c) Type I–E CRISPR; (d) Type I–C CRISPR

was used to deduce the genomic context of the CRISPR–Cas system [9].

2.3 | Analysis of direct repeats (DRs)

The RNAfold Webserver (<http://rna.tbi.univie.ac.at/cgi-bin/RNAWebSuite/RNAfold.cgi>) set to default parameters was used to predict the RNA secondary structure and minimum free energy (MFE) of each DR [10].

2.4 | Cas protein phylogeny

The Cas1 alignment was performed using the Multiple Sequence Comparison by Log-Expectation (MUSCLE) alignment algorithm and a maximum likelihood tree was generated in MEGA version 5 (<https://www.megasoftware.net/>) [11] with 1000 bootstrap replicates. The MEGA software is an integrated tool for conducting sequence alignment and inferring phylogenetic trees. The MUSCLE software used for protein alignment is freely available as a plugin in MEGA.

2.5 | Spacer analysis

The spacer targets were identified using the CRISPRminer2. The visual representation of the CRISPR spacers was performed using Excel macros, with each unique color combination representing one unique spacer sequence.

3 | RESULTS

3.1 | Confirmed CRISPR loci of *Aeromonas*

Overall, the genus *Aeromonas* showed a low frequency of CRISPR–Cas occurrence (10.66%, 13 of 122 genomes). The genus *Aeromonas* mostly harbors Type I CRISPR–Cas loci. Based on CRISPRminer2 prediction, most of the studied CRISPR–Cas systems were Type I–F (69.2%), followed by Type I–E (23.1%) and Type I–C (7.7%) (Table 1). The identified CRISPR–Cas systems and their features are listed in Table 1.

Analysis of the complete genomes of the *Aeromonas* revealed the CRISPR system to be more prevalent in *A. caviae* (22.7%) and *A. veronii* (20%), whereas only one strain of *A. hydrophila*, *A. salmonicida*, *A. schubertii*, and *A. simiae* possessed a complete Type I–F CRISPR–Cas system (Table 1). Among the five strains of *A. caviae* that

encoded the complete CRISPR loci, four were of the Type I–F system, and one of the Type I–E system. Similarly, in *A. veronii* strains harboring the complete CRISPR loci, the Type I–E system was observed in two strains, Type I–F and Type I–C in one strain each.

3.2 | Genomic context analysis of the confirmed CRISPR loci

Genomic context analysis of the CRISPR loci indicated that the Type I–F CRISPR system of *A. caviae* R25-6 was associated with transposition-related proteins and antibiotic resistance genes, such as the major facilitator superfamily (MFS) efflux pump FloR and aminoglycoside resistance genes (Figure 2). A similar arrangement of genes was evident in the genomic context of other strains such as *A. hydrophila* ZYAH75, *A. caviae* 1607-10029, and *A. caviae* WP2-W18-ESBL-01 (Supporting Information File S1). The Type I–F region of *A. veronii* strain JC529 was located close to proteins such as prophage protein and site-specific recombinase, which are encoded by mobile elements (Supporting Information File S1). The Type I–E CRISPR locus of *A. veronii* R25-2 was flanked by mobile element proteins and integrase (Supporting Information File S1).

3.3 | DRs

Analysis of the DR sequences indicated that they were mostly conserved in each CRISPR/Cas subtype. The DR length ranged from 28 to 29 base pairs. All the DRs formed stable RNA secondary structures with stems in the middle, as indicated by the MFE values (Figure 3). DRs 1, 2, and 6 had a stem length of 6 bp while DRs 3, 4, and 5 had a stem length of 7 bp (Figure 3). The MFE values indicated that DR3 and DR5 were more stable than DR1, DR2, DR4, and DR6 (Figure 3).

3.4 | Cas genes

Nine strains of *Aeromonas* that harbored CRISPR subtype I–F were associated with genes WYL, Cas1, Cas3f, Cas8f, Cas5f, Cas7f, and Cas6f (Figure 1a). In addition to these genes, the *A. caviae* 1607-10029 strain possessed the transposon associated TniQ, TnsC, and TnsB genes in the vicinity of the CRISPR locus (Figure 1b). The genes encoded by the Type I–E CRISPR loci of *Aeromonas* were WYL, Cas3, Cas8e, Cse2gr11, Cas6e, Cas7, Cas5, Cas1, and Cas2 (Figure 1c). However, the *A. caviae* R25-2 strain lacked the WYL and Cas2 genes. The CRISPR of

TABLE 1 Features of the CRISPR-Cas systems in completely sequenced genomes of the genus *Aeromonas*

Organism	Strain	Origin and year of isolation	Source	GenBank acc. no.	CRISPR-Cas type	Cas and associated proteins	No. of spacers
<i>Aeromonas hydrophila</i>	ZYAH75	China (2015)	Wound secretion	NZ_CP016990	IF	WYL, Cas1, Cas3f, Cas8f, Cas5f, Cas7f, Cas6f	51
<i>Aeromonas salmonicida</i>	O23A	Poland (2010)	Rock biofilm and bottom sediments	NZ_CP021654	IF	WYL, Cas1, Cas3f, Cas8f, Cas5f, Cas7f, Cas6f	54
<i>Aeromonas caviae</i>	WP2-W18-ESBL-01	Japan (2018)	Wastewater treatment plant effluent	AP021927	IF	Cas6f, Cas7f, Cas5f, Cas8f, Cas3f, Cas1, WYL	55
<i>A. caviae</i>	WP5-W18-ESBL-02	Japan (2018)	Wastewater treatment plant effluent	AP022110	IF	Cas6f, Cas7f, Cas5f, Cas8f, Cas3f, Cas1, WYL	48
<i>A. caviae</i>	R25-6	China (2016)	Sludge from bioreactor	NZ_CP025705	IF	Cas6f, Cas7f, Cas5f, Cas8f, Cas3f, Cas1, WYL	63
<i>A. caviae</i>	1607-10029	China (2016)	Shunt fluid from cholangitis patient	NZ_CP047981	IF	TniQ, TnsC, TnsB, Cas6f, Cas7f, Cas5f, Cas8f, Cas3f, Cas1, WYL	72
<i>A. caviae</i>	R25-2	China (2016)	Sludge from bioreactor	CP025777	IE	Cas1, Cas6e, Cas5, Cas7, Cse2gr11, Cas8e, Cas3	68
<i>Aeromonas veronii</i>	JC529	China (2014)	Fish pond	NZ_CP058912	IF	Cas6f, Cas7f, Cas5f, Cas8f, Cas3f, Cas1, WYL	61
<i>A. veronii</i>	17ISAe	South Korea (2017)	Discus spleen	NZ_CP028133	IE	WYL, Cas3, Cas8e, Cse2gr11, Cas6e, Cas7, Cas5, Cas1, Cas2	38
<i>A. veronii</i>	X11	China (2013)	Fish	NZ_CP024930	IE	WYL, Cas3, Cas8e, Cse2gr11, Cas7, Cas5, Cas6e, Cas1, Cas2	58
<i>A. veronii</i>	FC951	India (2017)	Human feces	CP032839	IC	Cas2, Cas1, Cas4, Cas7, Cas8e, Cas5, Cas3, WYL	53
<i>Aeromonas schubertii</i>	LF1708	China (2017)	<i>Oreochromis niloticus</i>	NZ_CP039611	IF	Cas6f, Cas7f, Cas5f, Cas8f, Cas3f, Cas3-Cas2, Cas1, WYL	104
<i>Aeromonas simiae</i>	A6	China (2017)	Sewage	NZ_CP040449	IF	WYL, cas1, cas3f, cas8f, cas5f, cas7f, cas6f	23

Abbreviation: CRISPR, clustered regularly interspaced short palindromic repeats.

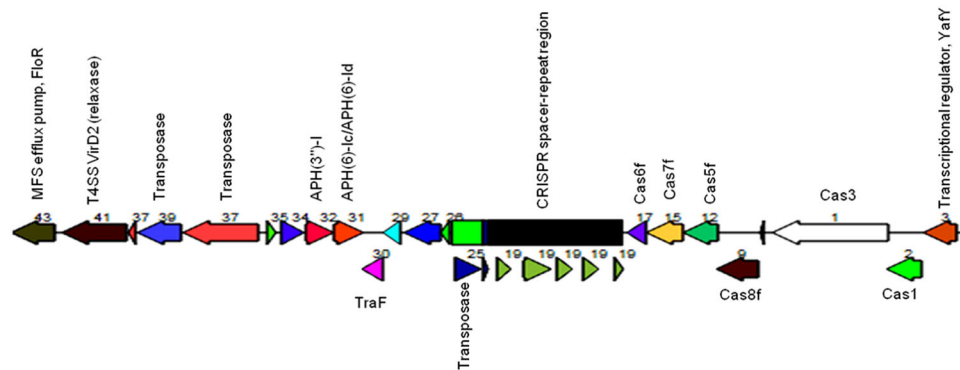
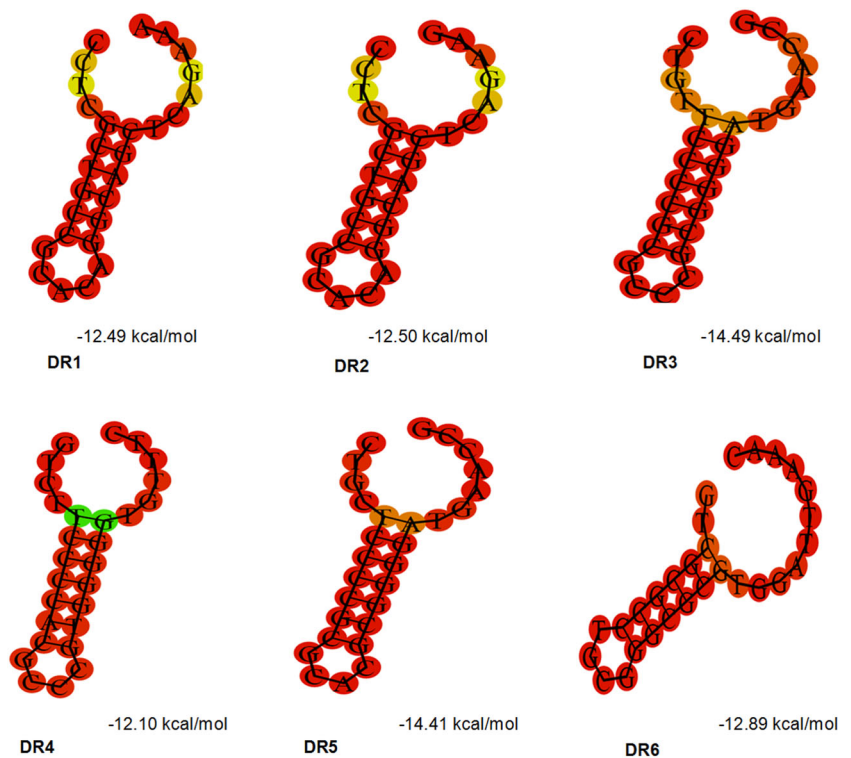


FIGURE 2 Genomic context of the Type I-F CRISPR-Cas system of *Aeromonas caviae* R25-6 visualized using the compare region viewer of the PATRIC database

FIGURE 3 RNA secondary structures and minimum free energy values of direct repeats (DRs) associated with the CRISPR loci of the genus *Aeromonas* as predicted by the RNAfold Webserver



A. veronii FC951 belongs to subtype I-C and encodes for Cas2, Cas1, Cas4, Cas7, Cas8c, Cas5, Cas3, and WYL (Figure 1d). A phylogenetic tree generated using the universal Cas1 protein to investigate the level of conservation or divergence between the identified CRISPR systems revealed the presence of three clusters that correspond to different CRISPR subtypes of the genus *Aeromonas* (Figure 4).

3.5 | Spacer analysis

In total, 748 spacers were identified among *Aeromonas* positive for CRISPR loci. Of the identified spacers, 627

were unique. The spacers in each array ranged from 23 to 104. Among *Aeromonas*, the least number of spacers (23) was seen within the CRISPR locus of *A. simiae*, while the highest number of spacers (104) was seen in *A. schubertii*. An analysis of spacer sequences showed 314 spacers to match plasmids (41.98%) and 129 spacers (17.25%) to match phages. The CRISPRminer2 prediction indicated the absence of self-targeting spacers. The spacer analysis indicated that the Chinese isolates *A. hydrophila* ZYAH75, *A. caviae* R25-6, *A. caviae* 1607-10029, and the Japanese isolate *A. caviae* WP2-W18-ESBL-01 which harbored the Type I-F CRISPR system had a common spacer repertoire. Furthermore, *A. schubertii* LF1708, *A. simiae* A6, and

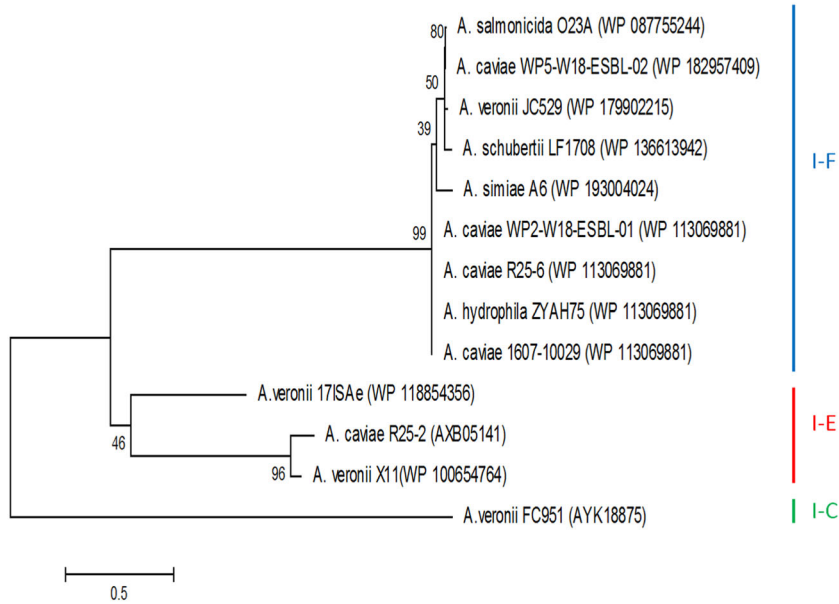


FIGURE 4 A phylogenetic tree generated using the MEGA software clustered the Cas1 protein into distinct groups. The CRISPR-Cas system type and subtypes are indicated on the right

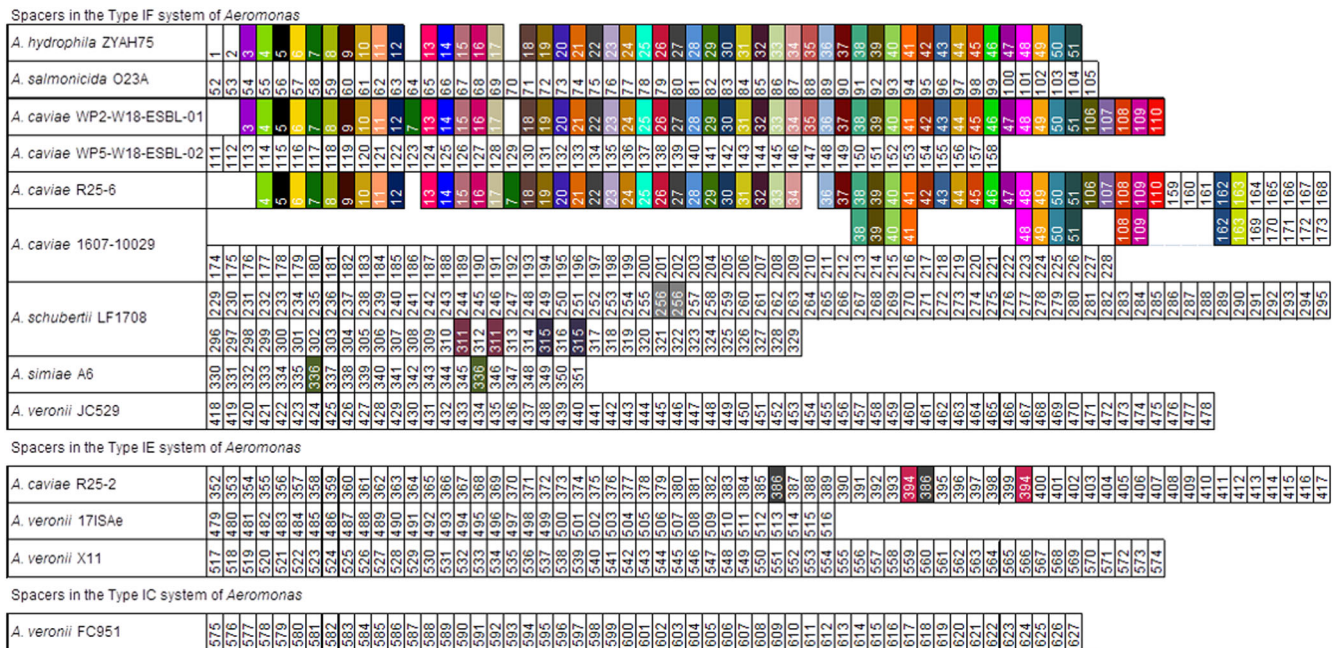


FIGURE 5 Graphical representation of the spacer organization in the different CRISPR–Cas subtypes of the genus *Aeromonas* visualized using Excel macros. The boxes with the same number and color represent identical spacers, while white represents unique spacers

A. caviae R25-2 (Figure 5) had duplicated spacers within the same CRISPR array.

4 | DISCUSSION

In this study, we investigated the CRISPR–Cas systems in the genus *Aeromonas* to gain insights into the occurrence, diversity, and features of its adaptive immune

system. The genus *Aeromonas* had a low frequency of occurrence of the complete CRISPR–Cas system (10.7%). This is comparable to the reported prevalence of complete CRISPR loci for *Escherichia coli* (9.4%) [12]. Further, it has been reported that only 10% of the major bacterial lineages host the CRISPR–Cas system, contrary to previous estimates of 81% and 40% occurrence in archaea and bacteria, respectively [13]. It has been hypothesized that the very low prevalence of CRISPR–Cas

systems in *A. veronii* is probably due to a lack of nucleotide biosynthesis capacity [14].

Based on the crRNA-effector complex, the CRISPR-Cas systems are classified into two general classes, Class I and Class II, which have been further categorized into 6 types and 33 subtypes [15]. The CRISPR-Cas system in members of the *Enterobacteriaceae* family mostly belongs to Class I and subtypes I-E and I-F [16].

In this study, a majority (69.2%) of the CRISPR-Cas systems of the *Aeromonas* species were of Type I-F and the Cas composition was similar to that of the CRISPR-Cas of *E. coli* and *Yersinia pestis* [17]. Further, in the Type I-F system, the *cas* genes are associated with Tn-7 bacterial transposon encoded genes [15]. However, in the Type I-F system of *Aeromonas*, except for one strain (*A. caviae* 1607-10029), no other species were associated with the transposition-related proteins (Table 1). The association of transposition-related proteins has been earlier reported in *Vibrio* spp. [18,19] and has been suggested to probably be involved in functions other than immunity [19,20].

Two strains of *A. veronii* and one strain of *A. caviae* in this study possessed the Type I-E CRISPR-Cas system. The phylogenetic analysis of the Cas proteins indicated that the Type I-E branch diverged further into two distinct subclusters that correspond to strains isolated from different geographic locations. Earlier studies showed that the *cas1* genes of *E. coli* were more divergent than those of *Proteus* [21]. A comparative analysis showed the architecture of this system to be similar to the Type I-E system of the *E. coli* K-12 strain composed of *cas3*, *cse1*, *cse2*, *cas7*, *cas5*, *cas6e*, *cas1*, *cas2*, and the CRISPR array [16]. Similarly, the Type I-C CRISPR system of an Indian isolate of *A. veronii* (Table 1) had an organization similar to that of the *Pseudomonas aeruginosa* I-C system. Further, in *P. aeruginosa*, the I-C system was always seen to be encoded by mobile genetic elements, which have been suggested to target phages, plasmids, transposons, and conjugative elements [22]. This assumes significance as in *A. veronii*, a phage integrase gene was seen to be located downstream of the Type I-C system, the exact role of which needs to be elucidated.

Most Type I-F systems of the *Aeromonas* species were closely associated with several antibiotic resistance genes, such as aminoglycoside phosphotransferases, efflux pumps, and beta-lactamases (Supporting Information File S1). Such observations have been reported in the CRISPR system of *Enterobacteriaceae* family members such as *Klebsiella pneumoniae* [23], *Salmonella* [24], and *E. coli* [25].

The CRISPR-cas systems of *Enterobacteriaceae* are regulated by multiple regulatory proteins like Lrp, H-NS,

CRP, LeuO, BaeR, and external signals like pH or quorum sensing [16]. In this study, the CRISPR systems of *Aeromonads* were associated with a transcriptionally regulated multiple ligand-binding protein WYL (named after a conserved amino acid W-Y-L sequence motif), which senses modified nucleotides and their derivatives generated during viral infection [26]. In *Synechocystis* sp., a WYL domain protein has been attributed to function as a negative regulator of the I-D CRISPR-Cas system [27]. Thus, the presence of the WYL domain associated with all *Aeromonas* CRISPR subtypes assumes significance as they could probably play an important role in the regulation of the CRISPR-Cas system.

Due to their partial palindromic nature, CRISPR repeats produce stable hairpin-like secondary structures. The stem-loop structures of DRs are thought to facilitate the interaction between the spacers and Cas proteins [28]. An investigation of the RNA secondary structures and their MFE values indicated that all DRs could form stable structures with $\Delta G < -10$ kcal mol⁻¹. DR3 and DR5 (29 bps in length) had lower MFE values in comparison to DR1, DR2, DR4, and DR6 (28 bp in length). Longer repeats with lower MFE form stable structures [21,29]. Reports suggest that active CRISPR arrays tend to be long due to the continuous acquisition of spacers [30]. In this study, the CRISPR loci of a few *Aeromonads* harbored up to 104 spacers, indicating an active system. Although bacterial CRISPR loci are known to harbor an average of 50 spacers per loci, bacteria like *Haliangium ochraceum* incorporate up to 587 spacers into a single array [31]. The average length of the spacers in the *Aeromonas* CRISPR system was 32 bp. The CRISPR system of *E. coli* consists of spacers with an average length of 31 bp [12]. CRISPR systems containing spacers of length >30 bp are more active than loci with shorter spacer lengths [12].

In this study, geographically distant strains such as the Chinese isolates *A. hydrophila* ZYAH75, *A. caviae* R25-6, and *A. caviae* 1607-10029 carried a CRISPR cassette with spacers similar to those of the Japanese origin strain *A. caviae* WP2-W8-ESBL-01. Such observations have been recorded earlier for the bacterium *Thermus* isolated from different continents [32]. The analysis of CRISPR cassettes in human gut metagenomes also indicates a common CRISPR core between geographically distant populations [33]. This probably indicates the presence of a common reservoir of phages and other plasmid-bearing strains that could spread among unrelated individuals residing in distant geographical locations [34].

A few spacers within the CRISPR loci of the genus *Aeromonas* were duplicated. This has been earlier

recorded in the bacterium *P. aeruginosa* and *E. coli* [35]. The exact significance of this is not known.

Further experimental investigations are needed to assess the functioning of the CRISPR–Cas system in this bacterium as several other additional roles, such as regulation of gene expression or silencing of some genes, have been associated with this system [36].

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

The authors declare that all data supporting the findings of this study are available within the article and its Supporting Information files.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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